

A Transfer Learning Method for Intelligent Fault Diagnosis from Laboratory Machines to Real-case Machines

Bin Yang, Yaguo Lei, Feng Jia, Saibo Xing
State Key Laboratory for Manufacturing Systems Engineering
Xi'an Jiaotong University
Xi'an 710049, China
E-mail: yaguo lei@mail.xjtu.edu.cn

Abstract—It is difficult to train a reliable intelligent fault diagnosis model for machines used in real cases (MURC) because there are not sufficient labeled data. However, we can easily simulate various faults in a laboratory, and the data from machines used in the laboratory (MUL) contain fault knowledge related to the data from MURC. Thus, it is possible to identify the health states of MURC by using related fault knowledge contained in the data from MUL. To achieve this purpose, a transfer learning method named convolutional adaptation network (CAN) is proposed in this paper. The proposed method first uses domain-shared convolutional neural network to extract features from the collected data. Second, the distribution discrepancy between the learned features of the data from MUL and MURC is reduced by minimizing multi-kernel maximum mean discrepancy. Finally, pseudo label learning is introduced to train domain-shared classifier by using unlabeled data from MURC. The proposed method is verified by a transfer learning case, in which the health states of locomotive bearings are identified by using the fault knowledge contained in the data from motor bearings used in a laboratory. The results show that CAN is able to effectively identify the health states of MURC with the help of the data from MUL.

Keywords—Intelligent fault diagnosis, transfer learning, multi-kernel maximum mean discrepancy, pseudo label learning.

I. INTRODUCTION

Machine learning has been widely used in the field of intelligent fault diagnosis of machines in recent years [1]. Especially, with the rapid development of machine learning, advanced deep learning methods are introduced to intelligent fault diagnosis and have made some achievements [2]-[4]. Although these achievements are able to identify the health states of machines automatically, a common assumption of most studies should be noted: the availability of massive labeled data. However, the opposite is true for intelligent fault diagnosis of machines used in real cases (MURC). Two main reasons may support this point. Firstly, MURC usually operate under the healthy condition, and the faults seldom happen [5]. Thus, the collected data are difficult to contain all the health states of MURC. Secondly, the types of faults during the long-term operation of MURC are unknown. It is unrealistic to label the data manually due to strong knowledge dependence. Thus,

the majority of data are unlabeled. For the above reasons, there are not sufficient labeled data from MURC to train a reliable intelligent fault diagnosis model by supervised learning.

Fortunately, we can simulate various faults in a laboratory, and the data from machines used in the laboratory (MUL) contain fault knowledge related to the data from MURC. Thus, it is possible to build an intelligent fault diagnosis model to identify the health states of MURC by using the related fault knowledge contained in the data from MUL. However, the data collected from MUL and MURC are respectively subject to substantially different distribution. Therefore, the intelligent fault diagnosis model just trained with the data from MUL does not work directly for MURC.

Transfer learning [6] provides a promising idea to solve this problem. The transfer learning tasks involve two types of datasets that are respectively drawn from the source and target domains. The source domain has sufficient labeled data, while the target domain contains a large number of unlabeled data that have related knowledge but different distribution with the data in the source domain. The intuitive idea behind transfer learning is to adapt the distribution and improve the learning of the predictive model in the target domain using the knowledge contained in the source domain. To achieve this purpose, feature-based transfer approaches are widely carried out, and a few related studies have been proposed in intelligent fault diagnosis of machines. Lu et al. [7] proposed a domain adaptation method for fault diagnosis of bearings based on transfer learning among varying operation conditions. Wen et al. [8] designed transfer learning based sparse autoencoders for fault diagnosis of bearings under different operation conditions. Zhang et al. [9] designed a convolutional neural network (CNN) with kernel dropout for fault diagnosis of bearings under different operation conditions.

According to the literature review, the previous studies suffer from two weaknesses as follows. Firstly, the previous studies are limited to fault diagnosis of the same machine based on transfer learning just from one operation condition to another. When handling a transfer learning task among different machines, such as intelligent fault diagnosis from laboratory bearings to locomotive bearings, the previous

This research was supported by National Natural Science Foundation of China (61673311), NSFC-Zhejiang Joint Fund for the Integration of Industrialization and Informatization (U1709208), and National Program for Support of Top-notch Young Professionals.

studies may present degraded performance. Secondly, the unlabeled data in the target domain cannot help to train the domain-shared classifier due to the lack of label information.

In order to overcome the above two weaknesses, a transfer learning method named convolutional adaptation network (CAN) is proposed in this paper. CAN focuses on intelligent fault diagnosis of MURC by using the fault knowledge contained in the data from MUL. In the proposed method, multi-kernel maximum mean discrepancy (MK-MMD) is used to adapt the distribution of the learned features of the data from MUL and MURC. In addition, unlabeled data from MURC are used to train the domain-shared classifier by pseudo label learning.

II. THE PROPOSED METHOD

A. Transfer Learning Tasks

Let $\mathcal{D}^s$ and $\mathcal{D}^t$ respectively represent the source and target domains. We denote the sample spaces $\mathcal{X}^s \in \mathcal{D}^s$ and $\mathcal{X}^t \in \mathcal{D}^t$. The data samples $\mathbf{x}^s \in \mathcal{X}^s$ and $\mathbf{x}^t \in \mathcal{X}^t$ are drawn from the source and target domains. We also denote the label space $\mathcal{Y} = \{1, 2, \dots, k\}$, which contains k kinds of health states.

In this paper, we mainly focus on intelligent fault diagnosis based on transfer learning from MUL to MURC. Given that the data collected from MUL and MURC are respectively subject to the marginal probability distribution $P(\mathbf{x}^s)$ and $Q(\mathbf{x}^t)$. The source and target domains are defined as follows.

1) The source domain $\mathcal{D}^s = \{\mathcal{X}^s, P(\mathbf{x}^s)\}$ contains n_s labeled samples $\mathcal{X}^s = \{\mathbf{x}_i^s, \mathbf{y}_i^s\}_{i=1}^{n_s}$ from MUL.

2) The target domain $\mathcal{D}^t = \{\mathcal{X}^t, Q(\mathbf{x}^t)\}$ contains n_t unlabeled samples $\mathcal{X}^t = \{\mathbf{x}_i^t\}_{i=1}^{n_t}$ from MURC.

3) The source domain should satisfy the fault knowledge need of the target domain, i.e., $\mathcal{Y}^t \subseteq \mathcal{Y}^s \subseteq \mathcal{Y}$, where $\mathcal{Y}^s$ and $\mathcal{Y}^t$ are label spaces in the source and target domains.

Denote a domain-shared classifier $h(\cdot)$ that is trained by supervised learning with labeled samples in the source domain. In the transfer learning tasks, $h(\cdot)$ cannot work directly for the target domain. We aim at building an intelligent fault diagnosis model that extracts cross-domain features subject to similar distribution, and finally forces the classifier $h(\cdot)$ to minimize classification error of samples in the target domain.

B. Convolutional Adaptation Network

CAN respectively extracts the features from the data in the source and target domains, and then reduces the distribution discrepancy of the learned cross-domain features. Its structure is shown in Fig. 1. The proposed method employs domain-shared convolutional neural network (DSCNN), distribution adaptation, and pseudo label learning to complete the transfer learning tasks. Each part is described as follows.

1) DSCNN. CNN serves as the basic framework of DSCNN to simultaneously extract the features from the data both in the source and target domains. DSCNN includes convolutional layers, pooling layers, and full-connected layers.

In the convolutional layers, the kernel $\mathbf{k}^l \in \mathbb{R}^{H \times L \times D}$ is used to convolve the input vectors of the layer $l-1$, where H is the height of the kernels, L and D are the length and depth of the kernels. The convolutional process is expressed as follows.

$$\mathbf{u}_i^{l,l} = \sigma(\mathbf{x}_i^{l-1,l} * \mathbf{k}^l + \mathbf{b}^{c,l}) \quad (1)$$

where $l = \{s, t\}$ is the indexes of the source domain and the target domain, $\sigma(\cdot)$ is the activation function, $\mathbf{b}^{c,l}$ is the bias of the l th layer, and $\mathbf{x}_i^{l-1,l}$ is the feature map of the previous layer $l-1$.

After obtaining the feature map $\mathbf{u}_i^{l,l}$, max pooling is conducted to reduce the dimensions of the input feature map. The pooled feature map is presented as follows.

$$(\mathbf{v}_i^{l,l})_{n,m} = \max\{\mathbf{u}_{(i),j,m}^{l,l} | s(n-1)+1 \leq j \leq sn, \forall \mathbf{u}_{(i),j,m}^{l,l} \in \mathbf{u}_i^{l,l}\} \quad (2)$$

where $\max\{\cdot\}$ returns the maximum value of a vector, s is the height of the non-overlapping segments, and $\mathbf{v}_i^{l,l}$ is the pooled feature map of the i th sample.

We can extract the high-level features relying on stacking the convolutional layers and pooling layers in turn. The high-level feature is flattened into the 1-dimensional (1D) vector, which is the input of full-connected layers. Similar to the multi-layer neural network, the output of the full-connected layer is shown as follows.

$$\mathbf{x}_i^{l,l} = \sigma(\mathbf{w}^l \mathbf{x}_i^{l-1,l} + \mathbf{b}^{f,l}) \quad (3)$$

where $\mathbf{w}^l$ and $\mathbf{b}^{f,l}$ are the weights and the bias of the l th full-connected layer respectively.

Specifically, the parameter set of DSCNN is shown in Table I.

TABLE I. PARAMETER SET OF DSCNN

Layers	Parameters	Activation function	Output size
Input	/	/	1200×1
C ₁	Kernels: 9×1×20	ReLU	1192×20
P ₁	s: 2×1	/	596×20
C ₂	Kernels: 7×1×20	ReLU	590×20
P ₂	s: 2×1	/	295×20
C ₃	Kernels: 5×1×20	ReLU	291×20
P ₃	s: 2×1	/	145×20
C ₄	Kernels: 3×1×20	ReLU	143×20
P ₄	s: 2×1	/	71×20
F ₁ /Flatten	/	/	1420
F ₂	Weights: 1420×256	ReLU	256
F ₃ /Classification	Weights: 256×4	Softmax	4

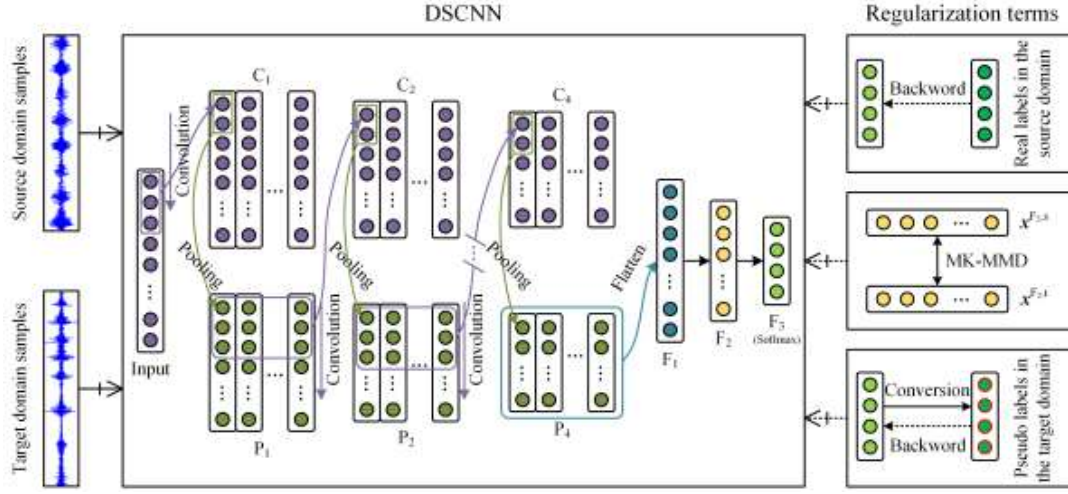


Fig. 1. Structure of the proposed CAN model.

2) Distribution adaptation based on MK-MMD. The learned cross-domain features are subject to different distributions. In order to reduce the distribution discrepancy, MK-MMD is first introduced to measure the discrepancy, and then the parameters of DSCNN are updated by minimizing MK-MMD.

It is considered that the distribution discrepancy of the learned cross-domain features is left in the full-connected layers. Moreover, the classification accuracy is related to the learned features in the layer F_2 . Thus, it is necessary to adapt the distribution of learned cross-domain features in the layer F_2 . Let $\mathcal{H}$ define the reproducing kernel Hilbert space (RKHS). Maximum mean discrepancy (MMD) [10] of the learned cross-domain features in the layer F_2 is calculated as follows.

$$D_{\mathcal{H}}^2(\mathcal{D}^s, \mathcal{D}^t) = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \Phi(\mathbf{x}_i^{F_2,s}) - \frac{1}{n_t} \sum_{j=1}^{n_t} \Phi(\mathbf{x}_j^{F_2,t}) \right\|_{\mathcal{H}}^2 \quad (4)$$

where $\Phi(\cdot)$ is the nonlinear mapping from the original feature space to RKHS. By means of the kernel mean embedding of distribution, RKHS is able to be induced by the characteristic kernels such as Laplace kernels and Gaussian kernels. Thus, the empirical estimation [10] of MMD based on the kernel mean embedding is calculated as follows.

$$D_{\mathcal{H}}^2(\mathcal{D}^s, \mathcal{D}^t) = \frac{1}{n_s^2} \sum_{i=1}^{n_s} \sum_{j=1}^{n_s} k(\mathbf{x}_i^{F_2,s}, \mathbf{x}_j^{F_2,s}) - \frac{2}{n_s n_t} \sum_{i=1}^{n_s} \sum_{j=1}^{n_t} k(\mathbf{x}_i^{F_2,s}, \mathbf{x}_j^{F_2,t}) + \frac{1}{n_t^2} \sum_{i=1}^{n_t} \sum_{j=1}^{n_t} k(\mathbf{x}_i^{F_2,t}, \mathbf{x}_j^{F_2,t}) \quad (5)$$

where $k(\cdot, \cdot)$ is the characteristic kernel. In this paper, multi-kernel characteristic kernel [11] is used for kernel selection, which is defined as follows.

$$\mathbf{K} := \left\{ \mathbf{k} : \mathbf{k} = \sum_{u=1}^d \beta_u \mathbf{k}_u, \sum_{u=1}^d \beta_u = 1, \beta_u \geq 0, \forall u \in \{1, \dots, d\} \right\} \quad (6)$$

where β_u is the parameter set for convex combination of d characteristic kernels. To solve the problem of convex quadratic program (CQP), β_u can be obtained by

$$\begin{aligned} \max_{\beta_u} \quad & \sum_{u=1}^d \beta_u \mathbf{k}_u / \sigma_u^2 \quad \text{where} \quad \sigma_u^2 = E[\beta_u \mathbf{k}_u]^2 - E^2[\beta_u \mathbf{k}_u] \\ \text{subject to} \quad & \sum_{u=1}^d \beta_u = 1, \beta_u \geq 0, \mathbf{k}_u \in \mathbf{K} \end{aligned} \quad (7)$$

Equation (5) indicates that the time complexity is $O(n^2 \times d)$, which leads to much time consuming during the training process of CAN. Thus, an unbiased estimation [10] of (5) is shown as follows.

$$\begin{aligned} \hat{D}_{\mathcal{H}}^2(\mathcal{D}^s, \mathcal{D}^t) = & \frac{1}{n} \sum_{i=1}^n k(\mathbf{x}_{2i-1}^{F_2,s}, \mathbf{x}_{2i}^{F_2,s}) + k(\mathbf{x}_{2i-1}^{F_2,t}, \mathbf{x}_{2i}^{F_2,t}) \\ & - \frac{1}{n} \sum_{i=1}^n k(\mathbf{x}_{2i-1}^{F_2,s}, \mathbf{x}_{2i}^{F_2,t}) + k(\mathbf{x}_{2i}^{F_2,s}, \mathbf{x}_{2i-1}^{F_2,t}) \end{aligned} \quad (8)$$

where $2n = n_s = n_t$. Equation (8) shows that the time complexity is reduced to $O(n \times d)$, which accelerates the training process effectively. According to (7) and (8), the distribution discrepancy of learned cross-domain features is reduced by the optimization objective as follows.

$$\min_{\theta^c} \max_{\mathbf{k} \in \mathbf{K}} \hat{D}_{\mathcal{H}}^2(\mathcal{D}^s, \mathcal{D}^t) \quad (9)$$

where θ^ℓ is the parameter set of DSCNN except the parameters of the layer F_3 .

3) Pseudo label learning. Although the distribution discrepancy of the learned cross-domain features is reduced by minimizing MK-MMD, the domain-shared classifier cannot be trained with the unlabeled samples in the target domain. Therefore, pseudo label learning is introduced. The pseudo label [12] of a sample is to pick up the label that has the maximum predicted probability as if it is the true label.

The generation of pseudo label includes probability prediction of labels and pseudo label conversion. In the layer F_3 , softmax function is used to predict the label of a sample, which is expressed as follows.

$$\mathbf{h}_i^{\mathbf{F}_3, I} = \left[P(y_i^I = 1 | \mathbf{x}_i^{\mathbf{F}_2, I}) \quad \dots \quad P(y_i^I = k | \mathbf{x}_i^{\mathbf{F}_2, I}) \right]^T, \quad (10)$$

$$P(y_i^I = j | \mathbf{x}_i^{\mathbf{F}_2, I}) = \frac{\exp(\left(\mathbf{w}_j^{\mathbf{F}_2}\right)^T \cdot \mathbf{x}_i^{\mathbf{F}_2, I} + \mathbf{b}^{\mathbf{F}_2})}{\sum_{j=1}^k \exp(\left(\mathbf{w}_j^{\mathbf{F}_2}\right)^T \cdot \mathbf{x}_i^{\mathbf{F}_2, I} + \mathbf{b}^{\mathbf{F}_2})}$$

where $\mathbf{h}_i^{\mathbf{F}_3, I}$ is the probability prediction of the labels for the i th sample, $\mathbf{x}_i^{\mathbf{F}_2, I}$ is the cross-domain features in the layer F_2 , $\mathbf{w}^{\mathbf{F}_2}$ and $\mathbf{b}^{\mathbf{F}_2}$ represent the weights and the bias of the layer F_2 .

According to (10), the pseudo label of the i th sample in the target domain is generated by

$$\hat{\mathbf{y}}_i^t = [\hat{y}_1^t \quad \hat{y}_2^t \quad \dots \quad \hat{y}_k^t], \quad (11)$$

$$\hat{y}_j^t = \begin{cases} 1 & \text{if } j = \arg \max_j \mathbf{h}_i^{\mathbf{F}_3, t} \\ 0 & \text{otherwise} \end{cases}$$

where $\hat{\mathbf{y}}_i^t$ is the pseudo label of the i th target domain sample.

The unlabeled samples in the target domain are assigned pseudo labels, and then DSCNN can be trained with unlabeled samples in the target domain by

$$\min_{\theta} \frac{1}{n_t} \sum_{i=1}^{n_t} \mathcal{L}(\hat{\mathbf{y}}_i^t, \mathbf{h}(\mathbf{x}_i^{\mathbf{F}_3, t})) \quad (12)$$

where $\mathcal{L}(\cdot, \cdot)$ is the cost function, θ is the parameter set of DSCNN.

C. Training Process

CAN is trained by jointly minimizing three regularization terms: 1) cross-entropy loss of the source domain samples with real labels, 2) MK-MMD of the learned cross-domain features in the layer F_2 , and 3) cross-entropy loss of the target domain samples with pseudo labels. Combined with (9) and (12), the optimized objective of CAN is presented as follows.

$$\min_{\theta} \max_{k \in \mathcal{K}} -\frac{1}{n} \sum_{i=1}^n (\mathbf{y}_i^s)^T \cdot \log(\mathbf{h}_i^{\mathbf{F}_3, s}) - \lambda \frac{1}{n} \sum_{i=1}^n (\hat{\mathbf{y}}_i^t)^T \cdot \log(\mathbf{h}_i^{\mathbf{F}_3, t}) + \alpha \hat{D}_{\mathcal{H}}^2(\mathcal{D}^s, \mathcal{D}^t) \quad (13)$$

where λ and α are tradeoff parameters to balance the contribution of regularization terms.

It is noted that Adam optimization algorithm is used in this paper to optimize (13). Specifically, the training process of CAN is detailed as follows.

Algorithm 1. Training process of CAN.

Input: Data samples $\{\mathbf{x}_i^s\}_{i=1}^{n_s}$ and $\{\mathbf{x}_i^t\}_{i=1}^{n_t}$, labels $\{\mathbf{y}_i^s\}_{i=1}^{n_s}$, parameters $\{\lambda, \alpha, \beta_u\}$.

Output: Parameter set θ and predicted labels of samples in the target domain.

Initialize parameters $\{\theta, \lambda, \alpha, \beta_u\}$.

for epoch $i = 1$ to N do

 for mini-batch $j = 1$ to n_s/n do

 #Forward propagate

 for layer $l = 1$ to $\ell - 1$ do

 Update β_u by CQP (7) and compute MK-MMD by (8).

 Generate pseudo label $\hat{\mathbf{y}}_j^t$ by (11).

 #Backward propagate

 for layer $l = \ell$ to 1 do

 Update parameters θ^l of the l th layer by Adam (13).

 Return parameter set θ and predict label for data $\{\mathbf{x}_i^t\}_{i=1}^{n_t}$.

III. TRANSFER LEARNING CASE: FAULT DIAGNOSIS FROM MOTOR BEARINGS TO LOCOMOTIVE BEARINGS

A. Dataset Description

The effectiveness of the proposed method is verified by a transfer learning case, in which the health states of locomotive bearings used in real cases are identified by using the fault knowledge contained in the data from motor bearings used in a laboratory.

The data from motor bearings used in the laboratory are provided by Case Western Reserve University. Single point faults of motor bearings were simulated by electro-discharge machining, and the fault diameter was 0.0014 inches. In the experiment, the sampling frequency was set as 12 kHz. The data were collected with a motor load 3 HP. As shown in Table II, the dataset A contains four kinds of health states including normal (N), inner race fault (IF), roller fault (RF), and outer race fault (OF). There are 404 samples, and each sample contains 1200 sampling points.

The data from locomotive bearings used in real cases are provided by the reference [13]. The test locomotive bearings have four health states, i.e., normal, rub faults of inner race, rub faults of roller, and rub faults of outer race. In the test, the sampling frequency was set as 12.8 kHz, and the load reached 9800 N. As shown in Table II, the dataset B has 404 samples, and each sample also has 1200 sampling points.

TABLE II. DATASET DESCRIPTION

Datasets	Bearing specs	Health states	No. of samples	Load
A	SKF6205	N	4×101	3 HP
		IF		
		RF		
		OF		
B	552732QT	N	4×101	9800 N
		IF		
		RF		
		OF		

According to Table II, we build the transfer learning task: A→B. The dataset A is viewed as the source domain to provide fault knowledge, while the dataset B is regarded as the target domain. The main purpose of the transfer learning task is to correctly classify the samples in the dataset B.

B. Transfer Results

In this section, the effectiveness of the proposed method is verified. The transfer results of CAN are compared with those of CNN and DDC. The framework of CNN is the same as CAN. CNN is first trained with the samples in the dataset A, and then the samples in the dataset B are classified directly. The framework of DDC is also the same as CAN, but there is no pseudo label leaning. For CAN, the tradeoff parameters λ and α are set as 0.05 and 1 respectively. Moreover, the characteristic kernel family used in MK-MMD is set as $[10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3]$. The training epoch is 100.

It is shown in Fig. 2 that CAN obtains the lowest training loss among the three methods, while CNN obtains the highest error during the training process. This result explains the effectiveness of CAN for distribution adaptation. For comparison, the training loss of DDC is lower than CNN but higher than CAN. This result proves that minimizing MK-MMD is able to reduce the distribution discrepancy of learned cross-domain features, meanwhile pseudo label learning helps to adapt the distribution of learned features effectively.

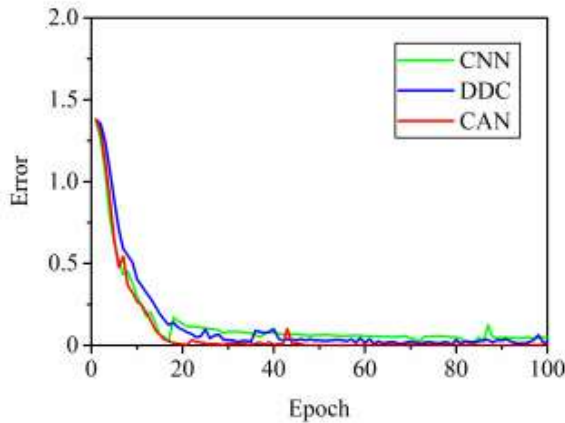


Fig. 2. Training loss of CNN, DDC, and CNN.

After training the three methods respectively, the samples in the dataset B are classified. It is noted that ten trials are implemented for each method and the statistics are shown in

Table III. It can be seen that CAN achieves the average accuracy of 88.86 % with a standard deviation of 2.07 %, which is the best one among the three methods. CNN obtains the lowest accuracy of 58.76 % with the highest standard deviation of 7.34 % because it cannot reduce the distribution discrepancy of learning cross-domain features. DDC obtains the accuracy of 85.01 % with a standard deviation of 3.26 %, which is better than CNN. This result proves that distribution adaptation using MK-MMD is able to achieve success in the transfer learning tasks. The classification accuracy of CAN is higher than DDC, which verifies the effectiveness of pseudo label leaning.

TABLE III. CLASSIFICATION ACCURACY (%) FOR DATASET B

Methods	Average accuracy	Standard deviation
CNN	58.76	7.34
DDC	85.01	3.26
CAN	88.86	2.07

In order to further understand the transfer learning process, the cross-domain features learned by CAN is presented via t-distributed stochastic neighbor embedding (t-SNE) algorithm. As shown in Fig. 3 (a), the learned features in the source and target domains suffer from distribution discrepancy. Thus, CNN cannot correctly classify the target domain samples when it is just trained with the source domain samples. As shown in Fig. 3 (b), DDC is able to reduce the distribution discrepancy of the learned features. However, some samples still suffer from serious domain discrepancy, such as the category of T-RF. Compared with the methods of CNN and DDN, CAN is able to further reduce the distribution discrepancy of the learned features in the source and target domains, which is shown in Fig. 3 (c). Therefore, the proposed method can reliably classify the target domain samples. This result visually gives the reason why CAN is able to achieve the transfer learning task.

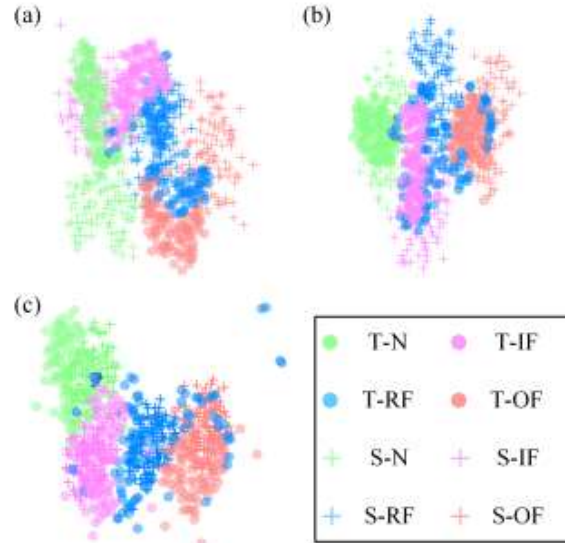


Fig. 3. Visualization of the learned features for dataset A (source domain) and dataset B (target domain): (a) CNN, (b) DDC, and (c) CAN

IV. CONCLUSION

This paper proposes CAN for intelligent fault diagnosis based on transfer learning from MUL to MURC. In the proposed method, the distribution adaptation based on MK-MMD and pseudo label learning is used to reduce the distribution discrepancy between the learned features of the data from MUL and MURC. The proposed method is validated by a transfer learning case of fault diagnosis from motor bearings used in a laboratory to locomotive bearings used in real cases. The results show that the learned features of the data from motor bearings and locomotive bearings are subject to the similar distribution. Therefore, the proposed method is able to identify the health states of locomotive bearings with the help of the data from motor bearings. The proposed method is not only limited to the transfer learning case of bearing fault diagnosis, but it can also be applied to the fault diagnosis of other machines, such as motors, gearboxes and hydraulic equipment. More transfer learning cases will be conducted in future work.

ACKNOWLEDGMENT

This research was supported by National Natural Science Foundation of China (61673311), NSFC-Zhejiang Joint Fund for the Integration of Industrialization and Informatization (U1709208), and National Program for Support of Top-notch Young Professionals.

REFERENCE

- [1] S. Yin, X. Li, H. Gao, and O. Kaynak, "Data-based techniques focused on modern industry: an overview," *IEEE Tran. Ind. Electron.*, vol. 62, no. 1, pp. 657-667, Feb. 2015.
- [2] Y. Lei, F. Jia, J. Lin, S. Xing, and S. X. Ding, "An intelligent fault diagnosis method using unsupervised feature learning towards mechanical big data," *IEEE Tran. Ind. Electron.*, vol. 63, no. 5, pp. 3137-3147, May. 2016.
- [3] H. Hu, B. Tang, X. Gong, W. Wei, and H. Wang, "Intelligent fault diagnosis of the high-speed train with big data based on deep neural networks," *IEEE Trans. Ind. Inform.*, vol. 13, no. 4, pp. 2106-2116, Aug. 2017.
- [4] F. Jia, Y. Lei, L. Guo, J. Lin, and S. Xing, "A neural network constructed by deep learning technique and its application to intelligent fault diagnosis of machines," *Neurocomputing*, vol. 272, pp. 619-628, Jan. 2018.
- [5] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, and J. Lin, "Machinery health prognostics: A systematic review from data acquisition to RUL prediction," *Mech. Syst. Signal Proc.*, vol. 104, pp. 799-834, May. 2018.
- [6] S. J. Pan, and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345-1359, Oct. 2010.
- [7] W. Lu, B. Liang, Y. Cheng, D. Meng, J. Yang, and T. Zhang, "Deep model based domain adaptation for fault diagnosis," *IEEE Tran. Ind. Electron.*, vol. 64, no. 3, pp. 2296-2305, Mar. 2017.
- [8] L. Wen, L. Gao, and X. Li, "A new deep transfer learning based on sparse auto-encoder for fault diagnosis," *IEEE Trans. Syst. Man Cybern.-Syst.*, to be published. doi: doi 10.1109/TSMC.2017.2754287.
- [9] W. Zhang, C. Li, G. Peng, Y. Chen, and Z. Zhang, "A deep convolutional neural network with new training methods for bearing fault diagnosis under noisy environment and different working load," *Mech. Syst. Signal Proc.*, vol. 100, pp. 439-453, Feb. 2018.
- [10] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schoelkopf, and A. Smola, "A kernel two-sample test," *J. Mach. Learn. Res.*, vol. 13, pp. 723-773, Mar. 2012.
- [11] A. Gretton, B. Sriperumbudur, D. Sejdinovic, H. Strathmann, S. Balakrishnan, M. Pontil, and F. Kenji, "Optimal kernel choice for large-scale two-sample tests," in *Advances in Neural Information Processing Systems*, Lake Tahoe, USA, Dec. 2012, pp. 1205-1213.
- [12] H. Lee, "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," in *International Conference on Machine Learning*, Atlanta, USA, Jun. 2013, pp. 2-8.
- [13] Y. Lei, Z. He, Y. Zi, and X. Chen, "New clustering algorithm-based fault diagnosis using compensation distance evaluation technique," *Mech. Syst. Signal Proc.*, vol. 22, no. 2, pp. 419-435, Feb. 2008.